



Education

University of Waterloo

BASc Biomedical Engineering - 86.31% Average / 3.7 GPA

Sep 2021 - Apr 2026

Waterloo, ON

Skills

- **Software:** C, C++, Python, Rust, MATLAB, FreeRTOS, AUTOSAR, BLE, I2C, SPI, UART, I2S, ADC
- **Hardware:** PCB Design, Schematic Capture, Altium, KiCad, LTSPICE, Signal Test, EMI/EMC, DFM, Soldering, Pick-and-Place
- **Test & Validation:** Oscilloscope, Logic Analyzer, DMM, STM32CubeMonPwr, CppUTest, Coverity, Hardware-in-the-Loop
- **Platforms & Tools:** ESP32, ESP-IDF, Raspberry Pi, Jetson, STM32, Git, Linux, SolidWorks, Jama, Eclipse, Jira

Experience

Electrical Engineer

OtO Inc.

Jan - Aug 2025

Toronto, ON

- Cut device sleep current **34%** and increased charging speed **47%** by integrating a new charging IC, shipped in all production units
- Owned **schematic capture** and **PCB layout** in **Altium**; hardware debug and test with oscilloscopes, logic analyzers, and DMMs
- **Hand-built** and validated PCBs with OpenPNP across **3 design iterations** ahead of successful **EMI/EMC** compliance testing
- Shipped ESP32 firmware features in C; I2C motor control, temperature-based charging, fail-safe battery handling, telemetry
- Built a **CI pipeline** with hardware-in-the-loop emulation, unit tests, linting, and static code analysis, catching firmware regressions

Product Hardware and Software Developer

Elev8 & Perform

May - Aug 2024

Toronto, ON

- Delivered working health prototype device from blank slate in **12 weeks**, owning R&D, BOM, resource management, and **demos**
- Designed PCB integrating ESP32 and gas/flow sensors; **3D-modeled** and printed device housing in SolidWorks across **4 iterations**
- Developed firmware in C with **FreeRTOS** scheduling, BLE telemetry, I2C sensor communications, capturing live user data
- Built a **Python** real-time visualization dashboard to monitor sensor data and device issues, enabling early-cycle hardware debugging
- Deployed a Flask REST API on Google Cloud with MongoDB, streaming live device telemetry into a web app for the user

Firmware Developer

Onsemi

Sep - Dec 2023

Waterloo, ON

- Shipped GPIO driver and HAL components in C for the RSL15 ARM Cortex-M33 SDK following **AUTOSAR** conventions
- Root-caused and resolved **customer-reported SDK bugs** by reproducing issues on hardware and patching sample-app code
- Authored CppUTest **unit-tests** and validated firmware releases with STM32CubeMonPwr, logic analyzers, and DMMs

Undergraduate Research Assistant

University of Waterloo - Department of Nanotechnology Engineering, Prof. Vivek Maheshwari

Jan 2024 - Jan 2025

Waterloo, ON

- Synthesized Au-Ca/Pt nanoparticle ERG electrodes for more accessible diagnostic strips, **outperforming** commercial products

Junior Systems Engineer

Ontario Health

Jan - Apr 2023

Toronto, ON

- Built a serverless medical record web app on Azure (Angular, Node.js, MongoDB), reducing **94%** infrastructure development cost

Projects

Embedded/Electrical Lead

VocalPoint - Capstone Project

Sep 2024 - Apr 2026

Waterloo, ON

- Led electrical and firmware architecture design for a Bluetooth microphone-array assistive device for hearing-aid users
- Designed a carrier PCB for Raspberry Pi CM5 and ESP32 in Altium, compactly integrating USB-C power, PCIe, I2S, and USB lines
- Built a BLE and I2C telemetry stack between ESP32 and Raspberry Pi CM5, enabling real-time control and audio data streaming

Electrical Lead

True North Biomedical Competition - Electrical Team

Sep 2025 - Apr 2026

Waterloo, ON

- Led 9-member electrical team to build an EMG sensor knee-brace; owned R&D, schematic capture, and firmware architecture
- Implemented analog EMG circuit (IA, active RC filters), and developed firmware (I2C, BLE, FFT) for the Teensy 4.1 and ESP32